Bui Le Duc Huy

Software Engineer

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Summary

Software Engineering graduate with hands-on experience in full-stack web development, RESTful APIs, and secure authentication/authorization systems. Proficient in TypeScript, C#, JavaScript, ASP.NET Core, Next.js, Node.js/Express, PostgreSQL, and SQL Server. Experienced in system architecture, test-driven development (TDD), and containerized deployments. Seeking a Software Engineer / Full-Stack Developer role to engineer scalable web applications and high-impact digital solutions.

Technical Skills

Programming: TypeScript, JavaScript (ES6+), C/C++, C#, Python
Web & API: Next.js, React.js, ASP.NET Core, Node.js/Express, RESTful APIs, EF Core, Prisma ORM
Database: PostgreSQL, SQL Server, SQLite, MongoDB
Testing & Tools: Playwright, Vitest, xUnit, Moq, Docker, Git/GitHub, AWS S3, Postman, axe-core, Better Auth, SSE
Embedded & IoT: ESP32, Raspberry Pi, Sensor/Actuator Integration, Embedded Systems

Project Experience

Mini Jira – Enterprise Agile Project Management Platform
Full-Stack Software Engineer | Personal Project | 08/2026 – 10/2026
Technologies: Next.js 16 (App Router), React 19, TypeScript, PostgreSQL, Prisma ORM 7, Better Auth, Tailwind CSS v4, Vitest, Playwright
GitHub: https://github.com/bhuy1313/mini-jira | Live Demo: https://mini-jira-neon.vercel.app

- Architected a production-grade Jira & Linear-inspired collaboration platform featuring multi-tenant workspaces, Scrum sprints, hierarchical sub-tasks, dynamic custom fields, and burndown analytics.
- Engineered an O(1) Fractional Decimal Positioning algorithm for Kanban drag-and-drop (@dnd-kit), completely eliminating PostgreSQL table-locking batch re-indexes under concurrent updates.
- Implemented server-authoritative Role-Based Access Control (RBAC) via a zero-trust Actor Context Pattern and engineered a real-time event pipeline using Server-Sent Events (SSE) with replay buffers.
- Developed a custom zero-dependency AST-based Markdown tokenizer and renderer with multi-word @mention autocomplete, eliminating XSS vulnerabilities by strictly avoiding dangerouslySetInnerHTML.
- Designed a 60 FPS GPU-accelerated micro-interaction and animation system (Linear-grade cubic-bezier curves, card hover elevations, and spring dialogs) fully compliant with WCAG 2.2 AA accessibility.
- Established a rigorous test pyramid achieving 500 automated Vitest unit/integration tests and 30+ Playwright E2E suites covering core journeys, visual diffs, and automated axe-core audits with 0 TypeScript errors.

ECommerce.Web – Modern Full-Stack E-Commerce Platform
Full-Stack Software Engineer | Personal Project | 05/2026 – 08/2026
Technologies: ASP.NET Core 10 MVC, C#, Entity Framework Core 10, SQL Server / SQLite, ASP.NET Core Identity, xUnit, Moq, Tailwind CSS, Docker
GitHub: https://github.com/bhuy1313/ECommerce.Web | Live Demo: https://ecommerce-demo-9phe.onrender.com

- Architected a production-ready e-commerce platform (.NET 10) featuring customer storefronts, address books, multi-tier checkout, coupon engines, and an administrative control portal.
- Integrated the VNPay online payment gateway with cryptographic HMAC-SHA512 checksum validation, return URL verification, and idempotent IPN webhook callbacks to prevent duplicate payment processing.
- Implemented verified-purchase product reviews, configurable shipping carrier methods, automated VAT tax computation, and promotional coupon algorithms (% and fixed discounts with total usage caps).
- Hardened application security utilizing ASP.NET Core Identity, Anti-Forgery Tokens (CSRF) across all mutating endpoints, brute-force account lockout (5 attempts / 5 mins), and production IP rate limiting.
- Built a clean, editorial minimalist design system (Anti-AI-Slop, neutral palette) with functional micro-interactions, container image zoom, and smooth cart transitions (@formkit/auto-animate).
- Established an automated test suite of 277 unit and integration tests (xUnit, Moq) verifying pricing calculations, inventory deductions, and security boundaries with containerized Docker deployment on Render.

Smart Power Control Box – IoT Embedded System
Developer | Team size: 4 | 12/2025 – 01/2026

- Developed an IoT-enabled power control system that activates electrical supply based on QR-code payment validation.
- Implemented embedded control logic on ESP32 to process transaction data, calculate usage time, and manage automatic power switching.
- Designed and assembled a custom PCB and integrated relays, TFT display, and peripheral components. Technologies: ESP32, Embedded C/C++, WiFi, PCB Design, Power Electronics

Additional Project

Parking Lot Counter & Sensor Display (Embedded/IoT Project)

- Built a vehicle entry/exit counter using push buttons and sensor-triggered logic.
- Displayed real-time parking occupancy and available slots on an LED display, including full/empty boundary handling.

SCZR – 2D Platformer Game (Unity, C#) | Team size: 2

- Developed core gameplay mechanics and character movement.
- Improved code structure and performance through reusable components.

Education

Major Software Engineering (2022 – 2026)
Van Lang University
GPA 3.17

Certificates

- 2023- 2024 RapidMiner Certificate in Data Analysis